

Math 118 – Fourier Analysis

Homework 3 — my solutions

Fall 2026 ■ Keshav Ramamurthy

HOMEWORK 3
due Wed, Sep. 23, 11:59 PM
(Upload your solutions to Gradescope)

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1. (4 points) (#28 p. 37.) Find the best-fit least squares parabola to the following data: $\begin{array}{c|cccc} x & 0 & 1 & 3 & 4 \\ \hline y & 0 & 8 & 8 & 20 \end{array}$
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2. (3 points) (#6 p. 35.) For $n > 0$, let

$$f_n(t) = \begin{cases} 1, & 0 \leq t \leq 1/n, \\ 0, & \text{otherwise.} \end{cases}$$

Show that $f_n \rightarrow 0$ in $L^2[0, 1]$. Show that f_n does *not* converge to zero uniformly on $[0, 1]$.

3. (3 points) (#7 p. 35.) For $n > 0$, let

$$f_n(t) = \begin{cases} \sqrt{n}, & 0 \leq t \leq 1/n^2, \\ 0, & \text{otherwise.} \end{cases}$$

Show that $f_n \rightarrow 0$ in $L^2[0, 1]$ but that $f_n(0)$ does not converge to zero.

4. (3 points) (#8 p. 35.) Is Theorem 0.10 true on an infinite interval such as $[0, \infty)$? Give a proof or counterexample. (Theorem 0.10 is on page 10: *If a sequence f_n converges uniformly to f as $n \rightarrow \infty$ on a finite interval $a \leq t \leq b$, then this sequence also converges to f in $L^2[a, b]$.*)
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5. (4 points) (#18 p. 36.) Suppose A is an $n \times n$ matrix with complex entries. Show that the following are equivalent:

- (a) The rows of A form an orthonormal basis in $\mathbb{C}^n$.
 - (b) $AA^* = I$ (the identity matrix).
 - (c) $\|Ax\| = \|x\|$ for all vectors $x \in \mathbb{C}^n$.
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6. (3 points) (#9 p. 84.) Expand the function $f(x) = \cos x$ in a Fourier sine series on the interval $0 \leq x \leq \pi$. Note: this means we expand $f(x) = \sum_{k=1}^{\infty} b_k \sin kx$, where $b_k = \frac{2}{\pi} \int_0^{\pi} f(x) \sin(kx) dx$. (See page 50 of the book.)
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Page numbers refer to “A First Course in Wavelets with Fourier Analysis, 2nd Edition” by Boggess and Narcowich.

Solutions

Problem 1

Solution. We are given a few data points:

$$x = 0, 1, 3, 4$$

with corresponding outputs $y = 0, 8, 8, 20$. In order to continue we consider an arbitrary polynomial

$$p(x) = ax^2 + bx + c$$

To proceed, let's calculate basic values of p that will help later:

$$p(0) = c, p(1) = a + b + c, p(3) = 9a + 3b + c, p(4) = 16a + 4b + c$$

Now, we have suppose the output vector for these inputs is

$$f = \begin{pmatrix} 0 \\ 8 \\ 8 \\ 20 \end{pmatrix}$$

And unlike a linear regression, our but now using a 4×3 matrix representing the the $x^2, x, 1$ terms.

$$Z = \begin{pmatrix} 0^2 & 0 & 1 \\ 1^2 & 1 & 1 \\ 3^2 & 3 & 1 \\ 4^2 & 4 & 1 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 \\ 8 \\ 8 \\ 20 \end{pmatrix}.$$

$$Z^T Y = Z Z^T \begin{pmatrix} a \\ b \\ c \end{pmatrix}.$$

Calculating $Z^T Y$ to start yields

$$Z^T Y = \begin{pmatrix} 0^2 & 1^2 & 3^2 & 4^2 \\ 0 & 1 & 3 & 4 \\ 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 8 \\ 8 \\ 20 \end{pmatrix} = \begin{pmatrix} 0^2(0) + 1^2(8) + 3^2(8) + 4^2(20) \\ 0(0) + 1(8) + 3(8) + 4(20) \\ 0 + 8 + 8 + 20 \end{pmatrix} = \begin{pmatrix} 400 \\ 112 \\ 36 \end{pmatrix}.$$

Now, let's calculate ZZ^T which is:

$$Z^T Z = \begin{pmatrix} 0^2 & 1^2 & 3^2 & 4^2 \\ 0 & 1 & 3 & 4 \\ 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0^2 & 0 & 1 \\ 1^2 & 1 & 1 \\ 3^2 & 3 & 1 \\ 4^2 & 4 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0^4 + 1^4 + 3^4 + 4^4 & 0^3 + 1^3 + 3^3 + 4^3 & 0^2 + 1^2 + 3^2 + 4^2 \\ 0^3 + 1^3 + 3^3 + 4^3 & 0^2 + 1^2 + 3^2 + 4^2 & 0 + 1 + 3 + 4 \\ 0^2 + 1^2 + 3^2 + 4^2 & 0 + 1 + 3 + 4 & 1 + 1 + 1 + 1 \end{pmatrix} = \begin{pmatrix} 338 & 92 & 26 \\ 92 & 26 & 8 \\ 26 & 8 & 4 \end{pmatrix}.$$

Now we can solve for constants a, b and c . We have :

$$\begin{pmatrix} 338 & 92 & 26 \\ 92 & 26 & 8 \\ 26 & 8 & 4 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 400 \\ 112 \\ 36 \end{pmatrix}.$$

This yields the constants $a = \frac{2}{3}, b = \frac{4}{3}, c = 2$ so our best fit quadratic is

$$\frac{2}{3}x^2 + \frac{4}{3}x + 2$$

□

Problem 2

Solution. In order to show that $f_n \rightarrow 0$ in $L^2[0, 1]$, Let's consider $f(k) = 0$, Then notice this proving this function goes to 0 is analogous to showing for some arbitrary $\epsilon > 0$,

$$\exists N \text{ such that } n > N \implies \|f_n\|_{L^2} = \sqrt{\langle f_n, f_n \rangle} < \epsilon$$

So this means we should calculate the inner product of f_n on itself, notice that is

$$\langle f_n, f_n \rangle = \int_0^{\frac{1}{n}} (1^2) dt + \int_{\frac{1}{n}}^1 0^2 dt = \frac{1}{n}$$

$$\text{So } \|f_n\| = \sqrt{\frac{1}{n}}$$

So when we consider the limit as $n \rightarrow \infty$, this is pretty trivial to show and can formulate it as follows:

For an arbitrary $\epsilon > 0$, we have

$$\exists N = \left\lceil \frac{1}{\epsilon^2} \right\rceil \text{ such that } n > N \implies \|f_n\| < \epsilon$$

so we are done. Let's prove that it does not uniformly converge, and this is pretty direct.

For a given $\epsilon > 0$, notice that we'd need to have

$$\|f_n\| < \epsilon, \forall t \in [0, 1]$$

But notice pretty straightforwardly if $t \in [0, \frac{1}{n})$ that $f_n(t) = 1$, and thus: $|f_n(t) - 0| \not< \epsilon$ for any $0 < \epsilon < 1$ so it can't be uniformly convergent $\square$

Problem 3

Solution. In order to do number 3, we can do the same as above. In order to determine L^2 convergence, consider an arbitrary $\epsilon > 0$, and we'd need to have

$$\exists N \in \mathbb{Z}^+ \text{ such that } n \geq N \implies \|f_n - f\|_{L^2} < \epsilon.$$

In this case, we are again examining convergence to 0, so let's consider

$$\|f_n - 0\|_{L^2} = \sqrt{\int_0^1 \langle f_n(t), f_n(t) \rangle dt}.$$

On $0 \leq t \leq \frac{1}{n^2}$, we have

$$\langle f_n(t), f_n(t) \rangle = \langle \sqrt{n}, \sqrt{n} \rangle = n,$$

so

$$\|f_n - 0\|_{L^2} = \sqrt{\int_0^{\frac{1}{n^2}} n dt + \int_{\frac{1}{n^2}}^1 0 dt}.$$

Evaluating the RHS integral gives

$$\int_0^{\frac{1}{n^2}} n dt = nt \Big|_0^{\frac{1}{n^2}} = \frac{1}{n}.$$

Thus,

$$\|f_n - 0\|_{L^2} = \sqrt{\frac{1}{n}} = \frac{1}{\sqrt{n}}.$$

Notice we can really easily bound this limit, and that means just setting

$$N = \left\lceil \frac{1}{\epsilon^2} \right\rceil + 1,$$

we have that

$$n \geq N \implies \|f_n - 0\|_{L^2} < \epsilon$$

and thus it converges to 0 in L^2 .

Now to show that $f_n(0)$ does not converge uniformly to zero, notice that

$$f_n(0) = \sqrt{n}.$$

Thus $f_n(0)$ (for non zero n) does not converge to 0, and $f_n(0) \rightarrow \infty$. $\square$

Problem 4

Solution. Let's try to recall normal uniform converge and then L^2 convergence on $[a, b]$ and then expand it to an infinite interval. We are given a function f_n that converges uniformly to f in this case we can write it, given an arbitrary $\epsilon > 0$.

$$\exists N \text{ such that } n \geq N \implies \|f_n(t) - f\| < \epsilon \quad \forall t \in \mathbb{R}^+$$

In contrast, the statement for L^2 convergence, can take the limit of the L^2 product on $[0, k]$ for $k \in \mathbb{R}^+$ and then take the limit as $k \rightarrow \infty$. This is

$$\lim_{k \rightarrow \infty} \int_0^k \|f_n(t) - f\|_{L^2}^2 dt.$$

However, notice that at any point $t \in \mathbb{R}^+$, we have that

$$\|f_n(t) - f\| < \epsilon \implies \|f_n(t) - f\|^2 < \epsilon^2$$

So we can bound this integral

$$\lim_{k \rightarrow \infty} \int_0^k \|f_n(t) - f\|_{L^2}^2 dt \leq \lim_{k \rightarrow \infty} \int_0^k \epsilon^2 = \lim_{k \rightarrow \infty} k\epsilon^2$$

Since $\lim_{k \rightarrow \infty} k\epsilon^2 = +\infty$ we are motivated to find a counterexample. We'll use a similar piecewise function in order to create something such that uniformly it converges to 0, but make it such that it has an L^2 inner product that doesn't converge to 0. To do so we consider

$$f_n(t) = \begin{cases} \frac{1}{n}, & 0 \leq t \leq n^2, \\ 0, & t > n^2. \end{cases}$$

In this function, uniform convergence is just for arbitrary $\epsilon > 0$, consider $N \geq \lceil \frac{1}{\epsilon} \rceil + 1$, and we'd then have that

$$n \geq N \implies \|f_n(t) - 0\| < \epsilon$$

so it uniformly converges to 0.

This is really convenient for us as when considering the L^2 inner product, notice that

$$\int_0^{n^2} \left(\frac{1}{n}\right)^2 dt + \int_{n^2}^{\infty} 0 dt = \frac{1}{n^2} \Big|_0^{n^2} = 1 \neq 0$$

so we have an effective counterexample. $\square$

Problem 5

Solution. Let's just do this by showing $(1) \implies (2) \implies (3) \implies (1)$

In order to do so, let's consider the column vectors a_i and we are given $\|a_i\| = 1, \langle a_i, a_j \rangle = 0$ by definition.

How does this show $AA^* = I$

Claim. $(A^*A)_{ii} = 1$ for $i \in \{1, 2, \dots, n\}$

Proof. Notice $(AA^*)_{ii}$ is a lot of computation, but we notice that if we can show A^* is a left inverse of A , since they are square matrices, we'd have

$$A^*A = I, \quad A^*A = I \implies AA^* = I.$$

And this is super nice because $(A^*A)_{ij}$ is just $\langle a_i, a_j \rangle$, and this is good because for an identity matrix, we just have to show

$$(A^*A)_{ii} = 1 \text{ and for } i \neq j, \text{ that } (A^*A)_{ij} = 0.$$

But notice that $(A^*A)_{ii} = \langle a_i, a_i \rangle = 1$ due to our vectors being normalized, and

$$(A^*A)_{ij} = \langle a_i, a_j \rangle = 0,$$

so we've shown $A^*A = I$, and we have shown $(1) \implies (2)$.

Now we'll show that $(2) \implies (3)$. This is pretty direct from the adjoint formula, namely:

$$\|Ax\|^2 = \langle Ax, Ax \rangle = \langle x, A^*Ax \rangle = \langle x, x \rangle = \|x\|^2.$$

Thus, $\|Ax\| = \|x\|$.

Now let's show that $(3) \implies (1)$.

In order to do so, we write $A = (a_1, a_2, \dots, a_n)$ for these column vectors, and similarly now denote $x = (x_1, x_2, \dots, x_n)$ for scalars $x_i \in \mathbb{C}$.

Notice here we have that:

$$\|x\|^2 = \sum_{i=1}^n |x_i|^2.$$

If we consider the value of the expression

$$\|Ax\| = \left\| \sum_{i=1}^n a_i x_i \right\|.$$

This sucks, so let's try phrasing it with the adjoint:

$$\langle Ax, Ax \rangle = \langle x, x \rangle \implies \langle x, A^*Ax \rangle = \langle x, x \rangle \implies \langle x, (A^*A - I)x \rangle = 0$$

for all $x \in \mathbb{C}^n$, or $A^*A = I$, which is (2). How does this show (1)?

We've shown so far $(1) \implies (2) \iff (3)$, and wish to show $(2) \implies (1)$.

In order for $A^*A = I$, we can use the same exact idea as showing $(1) \implies (2)$.

This is true only if $\langle a_i, a_i \rangle = 1$ and $\langle a_i, a_j \rangle = 0$ if $i \neq j$.

We claim that these conditions are sufficient for the a_i being an orthonormal basis in $\mathbb{C}^n$.

Let's first show that these are linearly independent. This would be true if $\sum_{i=1}^n c_i a_i = 0 \implies c_i = 0$ for constants c_i . We take the inner product of this with one arbitrary a_k , to get

$$\left\langle \sum_{i=1}^n c_i a_i, a_k \right\rangle = \langle 0, a_k \rangle \iff \sum_{i=1}^n c_i \langle a_i, a_k \rangle = 0.$$

This is really convenient as we've established when $i \neq k$, $\langle a_i, a_k \rangle = 0$, so this equation reduces to

$$c_k \langle a_k, a_k \rangle = c_k \cdot 1 = 0 \implies c_k = 0.$$

This inner product can be taken for each $k \in \{1, 2, \dots, n\}$, which then shows that each c_i must be 0, establishing linear independence. Along with this, the columns are already orthonormal as $\|a_i\|^2 = \langle a_i, a_i \rangle = 1$, so we are done (and it spans $\mathbb{C}^n$ since there are n linearly independent vectors) and we've shown

$$(1) \iff (2) \iff (3)$$

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Problem 6

Solution. We are given the Fourier sine series

$$f(x) = \sum_{k=1}^{\infty} \left(\frac{2}{\pi} \int_0^{\pi} \cos(x) \sin(kx) dx \right) \sin(kx).$$

Notice we can manipulate the integrand using

$$\cos(x) \sin(kx) = \frac{1}{2} (\sin((k+1)x) + \sin((k-1)x)).$$

Plugging this back into the sum, we get

$$f(x) = \sum_{k=1}^{\infty} \left(\frac{1}{\pi} \int_0^{\pi} (\sin((k+1)x) + \sin((k-1)x)) dx \right) \sin(kx).$$

Integrating, we get

$$f(x) = \sum_{k=1}^{\infty} \frac{1}{\pi} \left[-\frac{\cos((k+1)x)}{k+1} - \frac{\cos((k-1)x)}{k-1} \right]_0^{\pi} \sin(kx).$$

Evaluating this, for $k \neq 1$, this is equivalent to

$$b_k = \frac{1}{\pi} \left(1 - (-1)^{k+1} \right) \left(\frac{1}{k+1} + \frac{1}{k-1} \right)$$

And for the case $k = 1$, we have that $b_1 = 0$ because

$$b_1 = \frac{2}{\pi} \int_0^{\pi} \cos x \sin x = \frac{1}{\pi} \int_0^{\pi} \sin(2x) = \frac{1}{\pi} \left(-\frac{1}{2} \cos(2x) \right) \Big|_0^{\pi} = 0$$

For now we'll assume $k \neq 1$ and when k is odd, $k = 2k' + 1$, notice $(1 - (-1)^{2k'+1+1}) = 0$ so $b_k = b_{2k'+1} = 0$. Then, when k is even where $k = 2k'$, we have that this evaluates to

$$b_{2k'} = b_k = \frac{4k}{\pi(k+1)(k-1)}$$

So our final fourier series is

$$f(x) = \cos(x) = \sum_{k=1}^{\infty} b_k \sin(kx) = \sum_{k'=1}^{\infty} \frac{8k'}{\pi(2k'+1)(2k'-1)} \sin(2k'x) \text{ for } 0 < x < \pi$$

(where this is reindexed cause we only have non zero even indices)

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